

LLM Case Study Assignment

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Case Study 14: Resume Screening System

Introduction

Large Language Models (LLMs) are advanced AI systems based on the Transformer architecture. They understand natural language, extract information, summarize documents, perform reasoning, and interact with external systems through APIs. In enterprise applications, LLMs are combined with prompt engineering, fine-tuning, structured outputs, and security controls to deliver reliable results. In this section, introduction is discussed in detail with implementation considerations, practical examples, benefits, limitations, evaluation methods, and best practices. Organizations should validate outputs using business rules and human review where appropriate.

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Problem Statement

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LLM Architecture

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Pre-training

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Fine-tuning

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Transfer Learning

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Prompt Engineering

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Zero-shot Prompting

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One-shot Prompting

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Few-shot Prompting

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Reasoning-oriented Prompting

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Structured Prompting

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Suitability Score Generation

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LoRA / PEFT

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Advantages

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Conclusion

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Case Study 24: Function Calling for Banking

Introduction

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Pre-training

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Fine-tuning

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Transfer Learning

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Prompt Engineering

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Reasoning-oriented Prompting

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Structured Prompting

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Function Calling

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Functions

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Validation Requirements

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Security Considerations

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LoRA / PEFT

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LLM API Selection

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Prompt Evaluation

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Edge Cases

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Advantages

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Conclusion

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